

Anshuman Biswas

+91 7855043301 | anshuproductions@gmail.com | Gurgaon, Haryana | linkedin.com/in/anshuman-biswas-iiitk | github.com/Anshu666666

EDUCATION

Indian Institute of Information Technology, Kottayam

Bachelor of Technology in Computer Science and Engineering (CGPA: 9.08 / 10.0)

Kottayam, Kerala, India

Aug. 2024 – May 2028

Delhi Public School, Sushant Lok

Class XII (CBSE): 94.2% | Class X (CBSE): 97.2%

Gurgaon, Haryana, India

Apr. 2021 – May 2024

TECHNICAL SKILLS

Languages: C++, Python, TypeScript, JavaScript, SQL (PostgreSQL, MySQL), C, HTML/CSS

AI & Agent Architecture: LangGraph, LangChain, DeepAgents, Multi-Agent Systems, LLM Evaluation, RAG

Distributed Systems & Storage: LSM Trees, Write-Ahead Logs (WAL), Bloom Filters, Apache Kafka, Debezium, Redis, Elasticsearch

Backend & Cloud: FastAPI, Node.js, Express, REST APIs, WebSockets, SSE, Supabase, Cloudflare (R2/D1), GCP, Render

Developer Tools & Collaboration: Docker, Kubernetes, Linux, Git, GitHub Actions, CMake, Google Benchmark, ngrok, Jira, Slack

Spoken Languages: English (Fluent), Hindi (Fluent), Bengali (Native), French (Elementary)

EXPERIENCE

AI Engineering Intern

Oct. 2025 – Jul. 2026

theMonks.tech

Los Angeles, CA (Remote)

- Architected specialized multi-agent workflows with persistent memory, domain-specific execution skills, and user clarification loops to autonomously parse, transform, and plot complex datasets.
- Refactored sequential agent pipelines into dependency-aware DAGs, parallelizing independent tasks to achieve a **2× speedup** and cut overall response times from **30s to sub-15s**.
- Optimized container microservices to slash memory consumption by **80% (from ~250MB to ~50MB)**; benchmarked server deployments via **ngrok** tunnels, reducing cross-server latency by **85% (from 1.4s to 0.2s)**.
- Designed end-to-end system and application logging architectures; monitored live deployments and real-time telemetry across **GCP** and **Render** to track resource utilization, token budgets, and container health.

Marketing & Legal Intern

May 2026 – Present

ELSTRAT

Remote

- Conducted comprehensive market research and competitive analysis for early-stage technology startups across the Indian startup ecosystem.
- Assisted founding teams with legal documentation, regulatory compliance, and due diligence frameworks.

OPEN SOURCE CONTRIBUTIONS

Google Quantum AI (Cirq)

Sep. 2026

Core Developer Infrastructure & Release Pipeline Tooling

PR #8379 (Merged)

- Refactored developer release pipeline architecture and relative path hierarchies to standardize dev onboarding across the core repository.
- Completed Google Contributor License Agreement (CLA) verification and collaborated directly with Google Quantum AI software engineers (**Michael Hucka** and **Pavol Juhas**) through code review to immediate merge with 44/44 green CI test checks.

Munich Quantum Toolkit (TUM)

Sep. 2026

Quantum Algorithms & Compiler Benchmarking (MQT Bench)

PR #1037 (Merged)

- Engineered the scalable **Multipartite Superdense Coding** benchmark using n -qubit GHZ states (Bose et al., 1998; Hillebrand, 2012) to evaluate multi-qubit routing, compilation passes, and global entanglement distribution.
- Implemented local Pauli (X, Z) encoding logic with bit-flip propagation compensation ($c_i \oplus c_{n-1} = 1$) through the inverse-GHZ CNOT decoding cascade; isolated protocol stages with barriers to guarantee benchmark workload retention.
- Constructed exhaustive **StatevectorSampler** simulation test suites across all message permutations for $n \in \{2, 3, 4\}$ qubits with 100% test coverage; reviewed and merged by TUM researchers **Simon Hofmann** and **Daniel Haag**.

Tenue AI

Sep. 2026

Agent Capability Authorization & Offline Test Suite (CrewAI)

PR #716 (Merged)

- Architected the official **GuardedCrew** quickstart and offline test suite with a deterministic LLM double; validated role-based policies, argument regex constraints, and CrewAI 1.5–1.15 compatibility across 70 CI checks.

HOL-Guard (Hashgraph Online)

Sep. 2026

Static Analysis & Secret Scanner False-Positive Hardening

PR #3109 (Merged)

- Remediated false-positive credential alerts by engineering bracketed delimiter heuristics and regex boundary guards for template parameters (**#3091**); enforced fail-closed multiline tests.

MCP Memory Service

Sep. 2026

Security Hardening & Log Injection Mitigation

PR #1308 (Merged)

- Eliminated log injection vulnerabilities across configuration modules by converting eager f-strings to lazy parameterized logging and expanding **GUARDED_MODULES** regression coverage (14/14 CI checks passed).

PROJECTS

- TransactFlow** | *C++17, PostgreSQL, Apache Kafka, Debezium, Redis, Elasticsearch, Docker* Sept. 2025
– Engineered an enterprise Change Data Capture (**CDC**) streaming pipeline ingesting **1,000+ req/s** into PostgreSQL, decoding WAL with Debezium and replicating to dual sinks with **zero data loss** over 216k+ soak events.
– Developed high-throughput **C++** consumer workers featuring manual Kafka offset commits, exponential backoff retries, and Dead-Letter Queue (**DLQ**) isolation for poisonous payloads.
– Guaranteed sub-millisecond cache consistency by implementing atomic Lua-scripted Log Sequence Number (**LSN**) monotonicity guards in Redis.
- FlashDB** | *C++20, CMake, Google Benchmark, Python* Aug. 2025
– Designed and implemented an embedded Log-Structured Merge (**LSM**) tree key-value storage engine in **C++20** with decoupled Write-Ahead Log (**WAL**), MemTable, SSTables, and Size-Tiered Compaction.
– Integrated probabilistic **Bloom Filters** to eliminate redundant disk probes, achieving an **~85× read speedup** for non-existent key lookups under Google Benchmark stress testing.
- DeepTrade** | *Python, FastAPI, LangGraph, DeepAgents, Upstox API, Supabase, Telegram API* July 2025
– Architected a self-hosted agentic trading assistant coordinating specialized research and broker-execution subagents for real-time market queries and live/paper order execution on Upstox via Telegram with TOTP two-factor security.
– Investigated and upstream-reported two critical bugs in the official **Upstox Python SDK** ([#155](#), [#156](#)) regarding singleton configuration state leakage and urllib3 $\geq 2.0.0$ deprecations.

CERTIFICATIONS

- Anthropic:** Claude with the Anthropic API — Hands-on LLM integration, context-aware architectures, and agentic workflows (Cert: s2xodzvren54)
- LangChain Academy:** Project: Deep Agents — Building autonomous, long-running AI agents on LangGraph and LangChain (Cert ID: wtfjoejcfo)
- Oracle University:** Oracle Cloud Infrastructure (OCI) 2025 Certified AI Foundations Associate (Cert: 3231424800CI25AICFA)

ROLES OF RESPONSIBILITY

- BetaLabs (Official Technology Club of IIIT Kottayam)** Kottayam, Kerala, India
Lead – App Development Club | Sublead – AI & Data Science Club *Aug. 2025 – Present*
- **Lead, App Dev Club (Aug. 2026 – Present):** Core organizer and contributor for annual Techfest ”**Aaroh**” with 300+ participants; managed hackathons, technical workshops, and hands-on coding sessions primarily for freshmen and sophomore students.
- **Sublead, AI & DS Club (Aug. 2025 – Aug. 2026):** Conducted interactive hands-on sessions on foundational AI/ML concepts and volunteered in university-wide hackathons to mentor student development teams.